

AIML QUESTION PAPER

Que 1:

File Filtering. write all lines of a file1, except those that start with a pound sign (#), the comment character for Python to file2. And display data of file2.

Text file1 content:

Friends are crazy, Friends are naughty !

#Friends are honest, Friends are best !

Friends are like keygen, #friends are like license key !

We are nothing without friends, Life is not possible without friends !

Text file2 should be:

Friends are crazy, Friends are naughty !

Friends are like keygen,

We are nothing without friends, Life is not possible without friends !

Que 2:

Draw multiple plots in one figure using subplot function. The multiple plots include below according to order:

1. Plot a scatter plot with following data:

x = [5,7,8,7,2,17,2,9,4,11,12,9,6]

y = [99,86,87,88,111,86,103,87,94,78,77,85,86]

The x axis represents the age of car while y axis represents the speed of car. The title of the graph should be age v/s speed of car. Also in graph there should be x and y labels. The marker used should be star. The marker color should be green. The marker size should be 60.

2. Plot a horizontal bar with following data:

X = ["A", "B", "C", "D"]

Y = [3, 8, 1, 10]

The x axis represents the name of car while y axis represents the selling of car. The title of the graph should be name v/s selling of car. Also in graph there should be x and y label. The horizontal bar chart's height should be 0.1. The color of bar should be yellow.

3. Plot a histogram with following data:

Data = [1,3,3,3,3,9,9,5,4,4,8,8,8,6,7]

bins=4, the title of the graph should be histogram of cars. The orientation should be vertical. The color of plot should be violet

4. Plot a pie with following data:

Y = [35,25,25,15]

Mylabels = ['Apple','Bananas','Cherries','Dates']

The title of the graph should be pie chart. The exploded view should be shown with 0.2 value for 'Apple' . Also need to provide a superior title to the subplot prepared i.e 'My Subplot for cars' and subplot preparation

For clear visualization can use the following syntax after importing matplotlib

```
plt.figure(figsize=(10,10))
```

Que 3:

Using 'supermarket_sales.csv' file do the following operations and give required answer by using proper programming process.

- 1). Load the dataset into a pandas DataFrame and read first 8 rows.
- 2). Check for missing values and fill it by mean values of that particular column if any.
- 3). Find the number of orders which have 'Quantity' less than 3 and which have (either 'Rating' greater than 8.5 or 'Total' greater than 600).
- 4). Find the sum of 'Total' purchasing price spent by Member and Normal 'Customer type'.
- 5). Find the percentage of total of 'gross income' based on the different 'Payment' methods used by customers. (Ewallet, Cash and Credit card)
- 6). Analyze the purchasing behavior of male and female customers using 'Gender' column. Find their average purchase prices using 'Total' column.
- 7). Create a scatter plot that shows the relationship between total amount spent and rating. (keep '+' marker, with marker size 100 and green color).
- 8). Create a box plot that shows the distribution of 'Rating' and 'Quantity'. And comment about outliers in both columns.

Que 4:

Given the credit card transaction dataset, the objective is to build a classification model that accurately predicts whether a transaction is fraudulent or non-fraudulent based on the transaction features provided. The model should be trained on historical transaction data and evaluated on another portion of the dataset to assess its performance. The ultimate goal is to create a reliable classifier that can automatically detect fraudulent transactions and prevent financial losses for credit card companies and cardholders. Use `card_transdata.csv` for dataset.